

Animal behavior analysis in python

Day 3: Spatial Trajectories & Segmentation

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Outline

1. What is and Why Segment?
2. Overview of Segmentation Approaches
3. Classic Methods: Window, Displacement & Straightness
4. Change-Point Detection — Methodology
5. From Segments to Behavioral States
6. Afternoon Coding Session

1. What is segmentation?

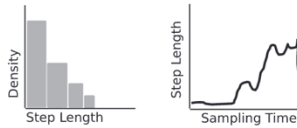
Actual Movement Path:



Step 1:



Step 2:



Step 3:



Step 4:

Cluster 1



Cluster 2

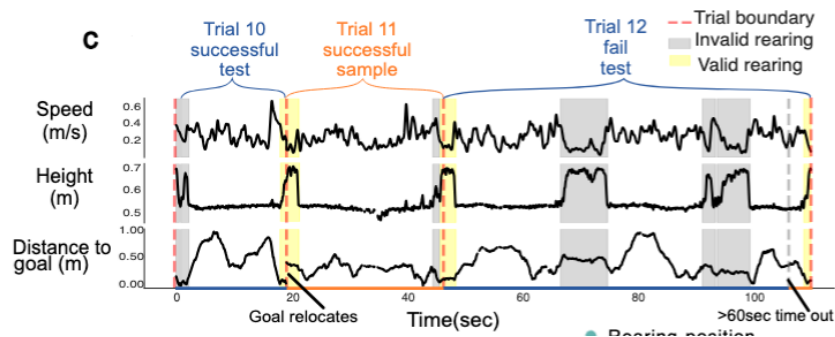
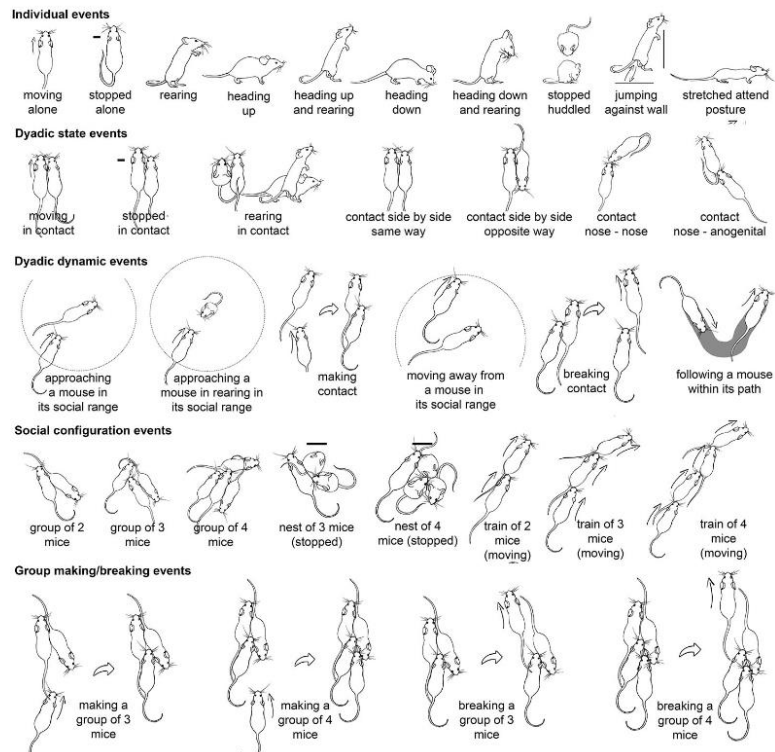


1. the actual continuous movement path of an organism is sampled as a set of consecutive relocations
2. path characteristics
3. distinguish different movement states

Edelhoff et al. 2016

2. Why Segment Trajectories?

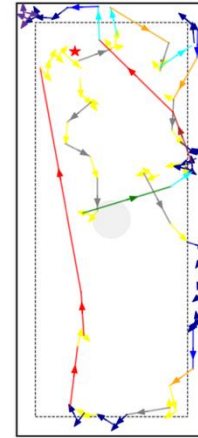
- Movement is inherently non-stationary



2. Why Segment Trajectories?

Segmenting trajectories helps isolate behaviors at biologically meaningful scales

distinguishing fine-scale area-restricted search from large-scale migratory movement



Biologically Meaningful Segments

Exploration: High speed, tortuous path, covering new ground

Local search: Low speed, tight loops near a location of interest

Resting / pausing: Near-zero velocity, stereotyped posture

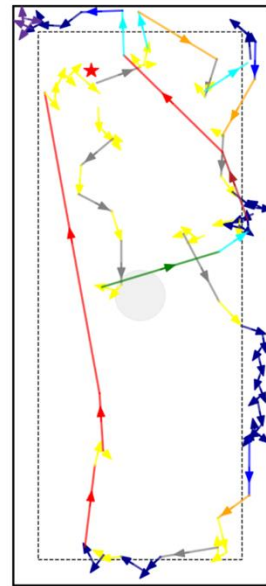
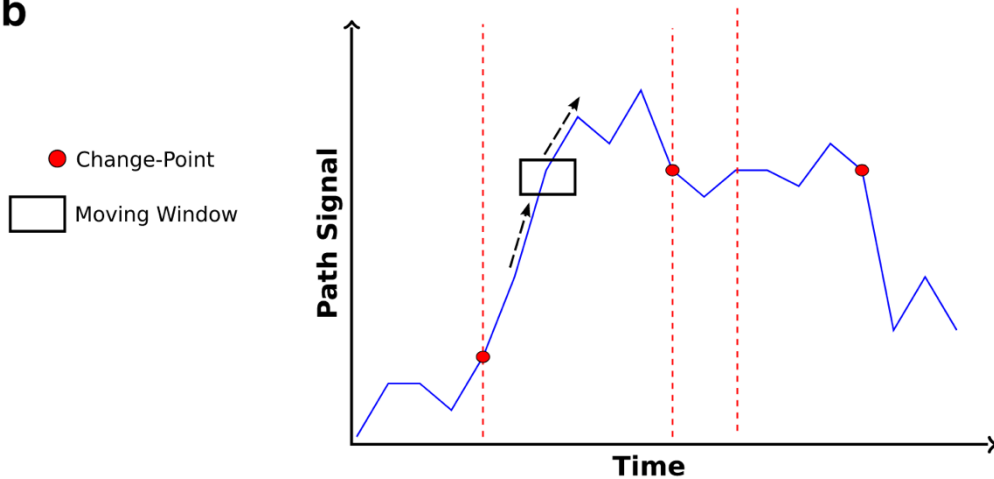
Directed travel: Straight, fast, persistent heading

2. Why Segment Trajectories?

Change-point detection

Segmentation identifies **where and when** behavioral transitions occur, then allow the detection of internal states

b

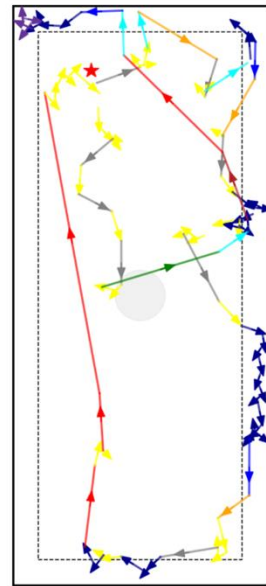
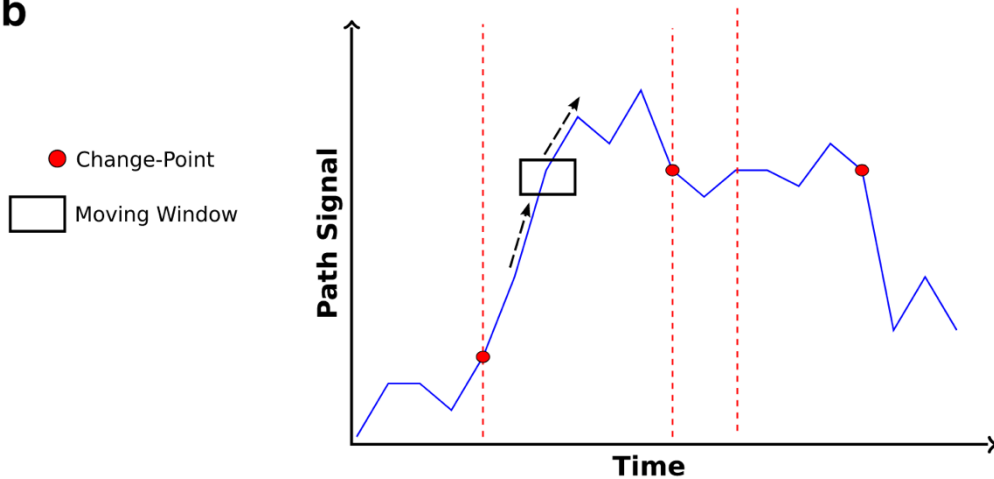


2. Why Segment Trajectories?

Change-point detection

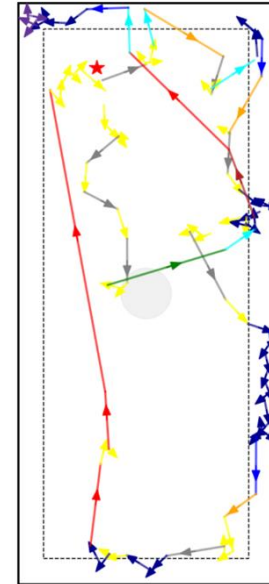
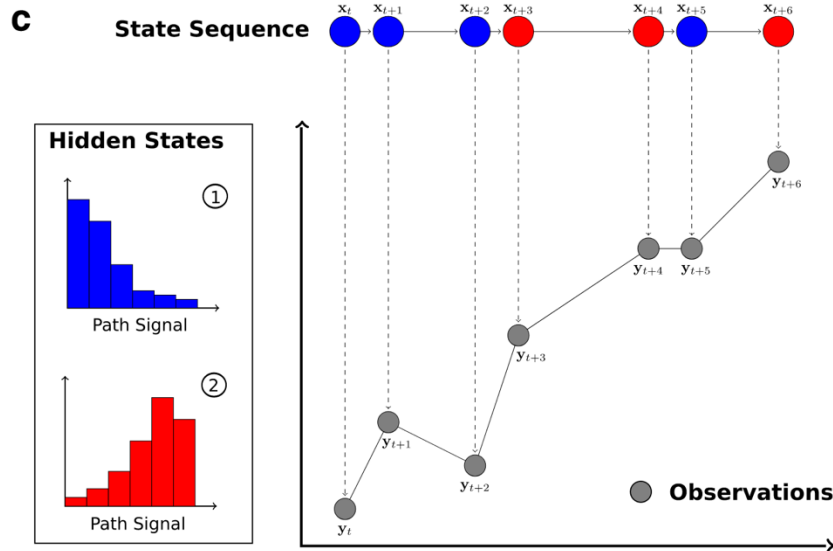
Segmentation identifies **where** and **when** behavioral transitions occur

b



2. Why Segment Trajectories?

Allow the detection of internal state



3. Segmentation features

Characteristic	Description	Type	Calculation
Displacement	Increment of the X and Y values between two consecutive relocations, change in absolute spatial position	primary	stepwise
Time lag	Duration / increment in time between consecutive relocations (usually determined by sampling regime)	primary	stepwise
Turning angles / heading	Relative and absolute turning angles between consecutive relocations, change in direction	primary	stepwise
Step length	Euclidean distance between two consecutive relocations	primary	stepwise
Velocity / speed	Distance traveled in a given time interval between two relocations; less sensitive to missing data than step length	primary	stepwise
Straightness index	Ratio of Euclidean distance between the beginning and end of a trajectory and the total path length (sum of all step lengths)	secondary	across multiple steps
Sinuosity / Tortuosity	Adaptions of the straightness index analyzing the probabilistic distributions of the changes in the turning angles and the beeline distance between the start and end points of the trajectory; index of path orientation	secondary	across multiple steps

3. Overview of Segmentation Approaches

A trajectory can be segmented at multiple levels of analysis. Three families of methods:

Window methods

Compute a local movement statistic (speed, angle, path length) in a sliding window

Threshold: more
subjectively

Geometric / displacement methods

Evaluate the shape of the path directly — straightness index, net displacement, or curvature within candidate segments.
Find the split that maximises within-segment homogeneity.

Statistical change-point detection

Model the trajectory signal as piecewise-stationary. Use statistical criteria (BIC, PELT, CUSUM) to locate the time points where the data distribution changes.

3a. Window-Based Methods

Compute local statistics in a sliding window

The Idea

For each frame t , compute a statistic over a local window $[t - w/2, t + w/2]$. Each window will be a segment

Common window statistics:

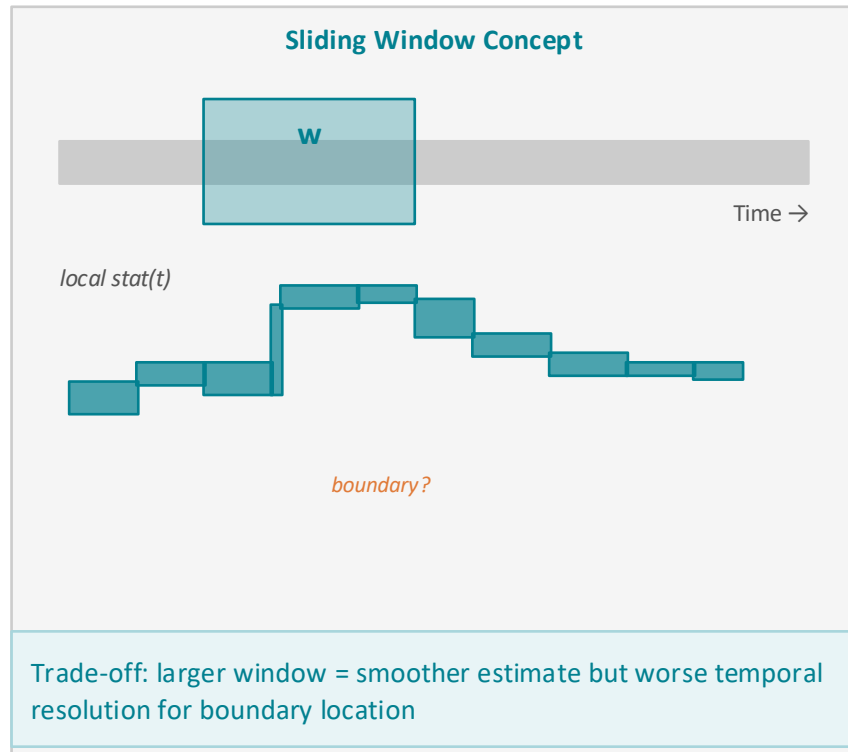
Mean speed: $\bar{v} = (1/w) \sum ||\Delta r|| / \Delta t$

Mean turning angle: $\bar{\vartheta} = (1/w) \sum |\alpha_i|$

Variance of heading: $\sigma^2_{\vartheta} = \text{Var}(\text{heading in window})$

Threshold detection:

Segment boundary time duration > threshold



3c. Straightness & Turning Angle Methods

Curvature as a fingerprint of behavioral state

Turning Angle (α)

The turning angle between consecutive steps i and $i+1$ is the change in heading direction.

$$\alpha_i = \arccos(\mathbf{v}_i \cdot \mathbf{v}_{i+1} / (|\mathbf{v}_i| |\mathbf{v}_{i+1}|))$$

$\alpha \approx 0^\circ$: Straight, directed movement

$\alpha \approx 180^\circ$: Reversal / U-turn

α variable, large: Tortuous local search

Compare net displacement to total path length

Straightness Index (SI)

For a segment from point i to point j , define:

$$SI = D_{\text{net}} / L_{\text{path}}$$

D_{net} = straight-line (Euclidean) distance from start to end

L_{path} = total path length (sum of step lengths)

$SI \in [0, 1]$ — $SI = 1$: perfectly straight | $SI \approx 0$: highly tortuous

Sinuosity Index

A related measure — Sinuosity (S) accounts for step length variability:

$$S = 1.18 \cdot \sigma_\alpha / \bar{v} \quad (\text{Benhamou 2004})$$

where σ_α = std of turning angles, $\bar{v}$ = mean step length

High S : tortuous path Low S : directed path

Practical Notes

- Smooth trajectory before computing angles — noise inflates turning angles
- Segment boundary: rolling window mean $|\alpha|$ crosses a threshold
- Both SI and α methods are fast, interpretable — good starting point

4. Change-Point Detection

Finding where a signal's statistical properties change

Given a signal $y_1, y_2, \dots, y_n$, find a set of change points $\{\tau_1, \tau_2, \dots, \tau_k\}$ such that the signal is approximately stationary (same distribution) within each segment, but the distribution changes across boundaries.

Optimization objective

PELT minimizes:

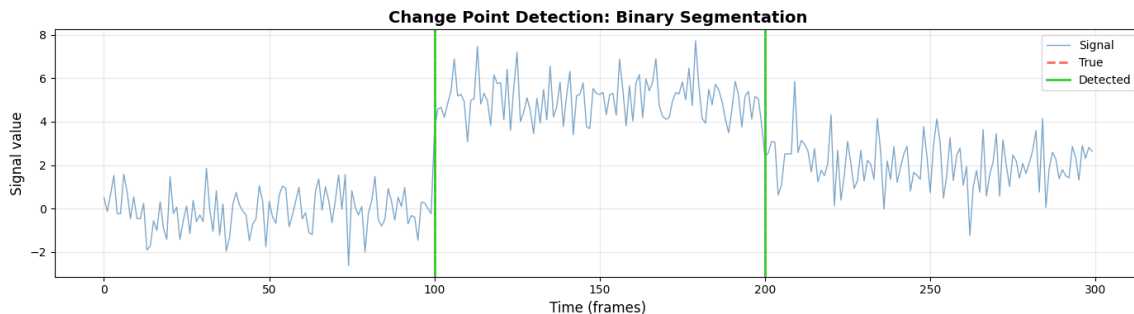
$$\sum_{\text{segments}} C(\text{segment}) + \beta \times (\text{number of changepoints})$$

Where:

- $C(\text{segment})$ = cost (e.g., variance or likelihood)
- β = penalty term controlling number of segments

PELT (Pruned Exact Linear Time)

Exact solution in $O(n)$ — best for long signals.
Default in ruptures library.



Binseg (Binary Segmentation)

Fast greedy approximation, $O(n \log n)$. Can miss multiple simultaneous changes.

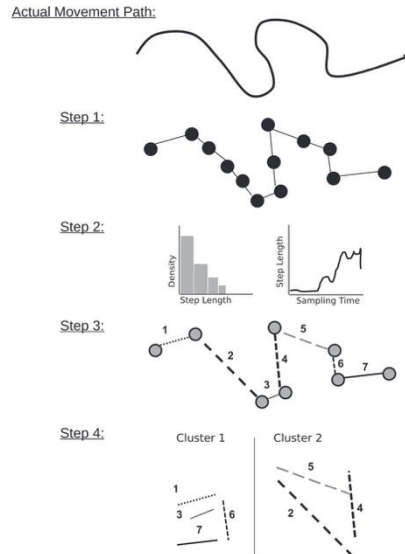
Full trajectory

↓ detect change
Segment1 | Segment2
↓ further splits ↓ further splits

5. From Segments to Behavioral States

Segmentation is a stepping stone, not the end goal

Analysis Pipeline



4. PELT — Pruned Exact Linear Time

The algorithm we will implement in the afternoon session

Core Idea

PELT finds the globally optimal set of change points using dynamic programming. It prunes candidate change points that can never be optimal — reducing runtime from $O(n^2)$ to $O(n)$.

Recurrence:

$$F(t) = \min_{s < t} [F(s) + C(y_{s+1:t}) + \beta]$$

Pruning Rule:

$$\text{Discard } \tau \text{ if: } F(\tau) + C(y_{\tau+1:t}) + \beta \geq F(t)$$

This candidate will never be optimal — remove it from future consideration.

Choosing the Cost Function $C(\cdot)$

L2 / Least Squares

Use when: Mean of signal changes

Signal: Speed, position

L1 / Median

Use when: Median changes (robust to outliers)

Signal: Angle, curvature

Normal (MLE)

Use when: Both mean AND variance change

Signal: Speed distribution

AR (autoregressive)

Use when: Temporal correlation structure changes

Signal: Acceleration

4. Choosing the Penalty β

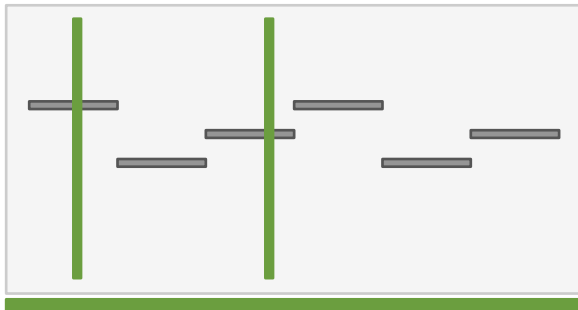
The most important hyperparameter in change-point detection

β controls the trade-off between fitting the data and model complexity (number of segments).



β too small → Over-segmentation

Too many spurious change points.
Noise is mistaken for real transitions.



β correct → Good segmentation

Change points match true behavioral transitions.
Segments are interpretable and consistent.



β too large → Under-segmentation

Real transitions are missed.
Behaviorally distinct segments are merged.

Systematic Penalty Selection

- BIC / AIC: $\beta = \log(n)$ (BIC) or $\beta = 2$ (AIC) — information criteria that balance fit and complexity
- Elbow method: Plot total cost $V(\beta)$ vs. number of segments — pick the inflection point ('elbow')
- Cross-validation: Withhold segments, predict, minimise prediction error — most principled but expensive